

SOMRAJ GAUTAM

Researcher in Document AI & Vision / Large-Language Models

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RESEARCH INTERESTS

Document Image Analysis • Table Understanding & Interpretation • Visual Question Answering • Active Learning • Multilingual Document AI • LLM / VLM • ASEAN Language Processing • NLP / NLU

EDUCATION

- Feb 2021 - Present **PhD in Computer Science and Engineering - IIT Jodhpur, India**
Advisor: Prof. Gaurav Harit
Thesis: Learning to See, Read, and Reason over Tables: A Unified Framework for Document Intelligence
Research Committee: Dr. V. Narayanan, Dr. Rajendra Nagar, Dr. Yashaswi Verma
- Aug 2018 - May 2020 **M.Tech in Computer Science and Engineering - IIT Dhanbad, India**
Advisor: Dr. Tanusree Kaibartta
Thesis: Test Cost Optimization of Core-Based Non-Stacked ICs
- Aug 2013 - May 2017 **B.Tech in Computer Science and Engineering - BIET Jhansi, India**
Advisor: Prof. A. K. Solanki

PUBLICATIONS

Peer-Reviewed Conference & Workshop Papers

- 1) **Somraj Gautam**, A. Bhandari, G. Harit. [TabComp: A Dataset for Visual Table Reading Comprehension](#). **NAACL 2025 (Core-A)**
OCR-free table QA benchmark with structured reasoning.
- 2) **Somraj Gautam**, N. Purohit, G. Harit. [Table Detection with Active Learning](#). **ICDAR 2025 (Core-A)**
Hybrid uncertainty + diversity sampling-based active learning for table detection.
- 3) **Somraj Gautam***, A. Subramanyam*, A. Bhandari, G. Harit. [Mind the \(Language\) Gap: Probing Numerical and Cross-Lingual Limits of LVLMs](#). **EMNLP 2025 Workshop MMCRICBENCH-3K: cross-lingual table VQA benchmark (Hindi/English)**.
- 4) **Somraj Gautam**, A. Dravichi, G. Harit. [INDOTABVQA: Cross-Lingual Table VQA on Real-World Bahasa Indonesian Document Images](#). **ACL 2026 (Core-A*)**
First real-world Indonesian cross-lingual table VQA dataset showing significant gains via fine-tuning and spatial priors.
- 5) **Somraj Gautam**, A. Bhandari, G. Harit. [Fragmented Financials: FinTab-Multi and the Challenge of Relational Reasoning Across Multiple Document Tables](#) **Under Submission EACL 2027 (Core-A)**
Benchmark + Method for robust reasoning over multiple financial table images.

- 6) **Somraj Gautam***, A. Dravichi*, G. Harit. *Beyond Semantic Similarity: Enforcing Bidirectional Entailment in Indonesian-English Code-Mixed Paraphrase Identification* **Under Preparation for TACL Journal**
- 7) **Somraj Gautam**, S.Khanna. *Can Vision-Language Models Trust What They Read? A Benchmark for Financial Table Integrity Verification* **Under Submission ECIR 2027 (Core-A)**

Journal Papers (Under Review)

- 1) **Somraj Gautam**, G. Harit. *Cycle-of-Thought: Self-Evaluative Iterative Reasoning for Table Visual Question Answering in Document Images*. **New Generation Computing (Under Review, IF: 2.3)**
Developed Self-RefineVQA, a training-free inference-time self-correction framework enabling VLMs to generate, critique, and iteratively refine reasoning for table-based VQA. Achieves consistent accuracy gains for small-scale models without supervised fine-tuning.

TEACHING & MENTORING EXPERIENCE

Feb 2021 - Present	Teaching Assistant - IIT Jodhpur
Aug 2021 - Present	Teaching Assistant - IIT(ISM) Dhanbad
May 2024 - May 2025	Mentored 1 MTech Student, which resulted in published work in ICDAR 2025
May 2025 - May 2026	Mentoring BTech Student, submitted to ACL 2026, and preparing for EMNL 2026.

RESEARCH & WORK EXPERIENCE

Feb 2021 - Present	Junior/Senior Research Fellow (JRF/SRF) - IIT Jodhpur Conducting PhD research on table interpretation in document images. Developing novel frameworks for table detection, table QA, active learning, and multilingual document understanding. Proposed Self-RefineVQA and contributed to multiple top-tier publications.
Jan 2021 - Sep 2021	Project Developer - National Informatics Center (NIC), Dantewada Developed Sanket, a Sign Language to Speech Converter using InceptionV3 to improve accessibility for the hearing-impaired community.

AWARDS & HONOURS

- Best Research Presentation - TLS Workshop 2025, IIT Delhi.
- Travel Grant - Summer School on Generative AI for Text, IIT Gandhinagar, 2024.
- Travel Grant - ACM India Winter School on Optimization for ML & Operations Research, 2022.
- Travel Grant - 6th Summer School on AI (Computer Vision & ML), IIIT Hyderabad.
- Ministry of Education (MoE) Fellowship Awardee - IIT Jodhpur, 2021.
- Ministry of Education (MoE) Fellowship Awardee - IIT(ISM) Dhanbad, 2018.
- Prime Minister's Scholarship Scheme (PMSS), Ministry of Defense, Government of India, 2013.
- International Mathematics Olympiad - Rank 4, 2010.

ACADEMIC SERVICE

Reviewer:

- ACL 2026
- NeurIPS 2026
- Journal: SN Computer Science (Springer).

Presentations

- S. Gautam et al.: An Approach of Bin Packing Technique for Test Cost Optimization of Core-Based Non-Stacked ICs. Workshop on RTL and High-Level Testing (WRTLTL-19), Asian Test Symposium, Kolkata, India, 2019. (Oral Presentation).

TECHNICAL SKILLS

Languages	Python, C++
ML Frameworks	PyTorch, TensorFlow, scikit-learn
Research Tools	HuggingFace Transformers, Weights & Biases, OpenCV
Infrastructure	Google Colab, Kaggle, Git, Overleaf

NATIONAL EXAMINATIONS & QUALIFICATIONS

- GATE 2020 (Computer Science)
- GATE 2018 (Computer Science)
- 3rd Rank - National Scholarship Test (NST), Made-Easy, 2017
- JEE Advanced 2013 | JEE Mains 2013

REFERENCES

1. Prof. Gaurav Harit
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2. Dr. Yashaswi Verma
Associate Professor
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